STRUCTURAL ECOSYSTEMIC TUNNELING HYPOTHESIS

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Epistemic status: Conceptual hypothesis open to empirical testing

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Abstract

It is proposed that certain observable harms affecting biophysical, human, technological, and network components may manifest an earlier loss of margin, distributed and interconnected through dependency relationships. The coexistence of impacts does not demonstrate a common cause. The hypothesis examines whether pressure-reduction capacities and corrective mechanisms sustainably exceed the generation of new pressures, and whether this results in verifiable material recovery.

Keywords: structural ecosystemic tunneling; margin; dependency; hyperacceleration; aggregate pressure; verifiable return; BIOIA/WUOM.

1. Starting point

BIOIA/WUOM records impacts on climate, territory, biodiversity, and human, technological, and network systems. The simultaneity of harms does not prove that they share a single cause or that propagation occurs between them. That possibility requires independent testing.

2. MICRO · Observed

At the biophysical foundation, climate alterations, territorial degradation, biodiversity loss, and transgressions of planetary boundaries are documented; in other components, costs, disruptions, and vulnerabilities emerge. Global climate indicators document continuing anthropogenic warming and Earth's energy imbalance, providing an observable biophysical context for examining loss of margin [4].

The olive fruit fly offers an analogy: the visible hole may appear after an internal tunnel has formed. The analogy is neither causal proof nor a biological model that can automatically be extrapolated.

3. MACRO · Structural hypothesis

Some visible harms may be delayed manifestations of an earlier loss of margin, distributed and connected through dependencies. Sequence to investigate: **loss of margin → internal or distributed deterioration → transmission through dependency relationships → visible manifestation → detection → accounted cost**. Each transition is a testable proposition, not an assumed fact.

4. Operational definition of margin

“Margin” refers to a system's distance from, or capacity relative to, specified thresholds and operating conditions, including its resilience, reserves, and ability to absorb disturbances without losing relevant functions. There is no

universal unit of margin: each study must identify the system, its historical reference, indicator, threshold, spatial scale, and time period. Recovery of one indicator does not necessarily amount to comprehensive restoration.

5. The foundation does not remain confined to the foundation

Earth provides the biophysical conditions necessary for the functioning of the whole. A loss of margin in that foundation can affect human, technological, and institutional systems through material dependencies. No automatic, exclusive, or linear causality is postulated: mechanisms, temporal sequences, magnitudes, and alternative explanations must be verified.

6. A change in mapping

Mapping isolated impacts records their distribution and accumulation. The hypothesis adds the possibility of internal connections: several harms may be different outputs of a shared deterioration process. Temporal coincidence alone is insufficient to support that interpretation; specific links and independent evidence of propagation must be identified.

7. Hyperacceleration and differential amplification

Inherited loss of margin, continuing pressures, hyperacceleration, and new technological capacities may coexist. The Montreal Protocol provides a precedent for the effective correction of a specific pressure and must remain a comparison case rather than a dismissed exception [5].

Efficiency per operation may improve while aggregate pressure increases. The International Energy Agency reported approximate increases of 17% in global data-centre electricity consumption in 2025 and 50% for AI-focused centres [1]. These figures illustrate a divergence between efficiency and scale; they do not, by themselves, demonstrate structural tunneling.

Corporate environmental reports allow shifts between accounting scopes to be studied. For fiscal year 2024, compared with its 2020 baseline, Microsoft reported a decline in Scope 1 and 2 emissions and an increase in Scope 3 emissions, alongside an increase in total emissions [3]. Inventory boundaries, methodological changes, and the comparison period must remain explicit. Google's calculations of “avoided” emissions [2] are counterfactual estimates, not equivalent reductions in absolute emissions.

8. Orientation variables: R and P

R = capacity to reduce or remove pressure. P = capacity to generate new pressure. These are orientation variables, not a physical equation or automatically comparable quantities. The question is whether, within explicit system boundaries and metrics, sustained net reduction enables verifiable recovery of margin. Local reduction must be distinguished from the displacement of pressures to infrastructure, materials, supply chains, or other territories.

9. Conditions for testing and possible refutation

The hypothesis gains support if, in delimited cases, an earlier loss of margin, a dependency chain transmitting deterioration, and subsequent manifestations compatible with that mechanism are demonstrated, while corrective measures fail to recover margin or displace pressures. Persistence of aggregate pressure alone is insufficient.

The hypothesis is weakened if the proposed propagation does not appear when mechanisms and temporal sequences are tested; if associations disappear after controlling for alternative factors; or if sustained corrections reduce the relevant

pressures and restore margin without significant displacement. A counterexample limits the hypothesis's scope, although it does not automatically invalidate every other case.

10. Temporality and return

Evaluation must connect three times: **Where do we come from?** (inherited conditions); **Where are we?** (state and response); **What future are we moving toward?** (trajectory). Reducing emissions, returning to a reference temperature, and restoring ecosystemic habitability are not equivalent outcomes. Return requires materially grounded, sustained evidence over time, acknowledging potentially irreversible losses.

11. Minimum evaluation protocol

• Delimit the system, scale, period, indicators, thresholds, and historical references.

• Document primary sources and distinguish measured data, estimates, and projections.

• Identify dependency mechanisms and establish the temporal order of changes.

• Test alternative explanations and cases in which recovery did occur.

• Compare total pressure, unit efficiency, displaced effects, and recovered margin.

• State uncertainty, limitations, and conditions under which the hypothesis would be weakened.

12. Epistemic status

HYPOTHESIS · NOT DEMONSTRATED. MICRO = observed; MACRO = inferred; RESULT = open. The cited evidence documents phenomena compatible with components of the hypothesis; it does not demonstrate a universal ecosystemic mechanism. The absence of an external response or refutation does not constitute validation. Independent testing and a return to the territory are invited.

References

[1] International Energy Agency (2026). *Key Questions on Energy and AI*. https://www.iea.org/reports/key-questions-on-energy-and-ai

[2] Google (2026). *2026 Environmental Report*. https://sustainability.google/google-2026-environmental-report/

[3] Microsoft (2025). *2025 Environmental Sustainability Report*. https://www.microsoft.com/en-us/corporate-responsibility/sustainability/report

[4] Forster, P. M., et al. (2026). *Indicators of Global Climate Change 2025: annual update of key indicators of the state of the climate system and human influence*. *Earth System Science Data*, 18, 3889–3933. https://doi.org/10.5194/essd-18-3889-2026

[5] UNEP, Ozone Secretariat. *Facts and figures on ozone protection*. https://ozone.unep.org/facts-and-figures-ozone-protection